

Most economic observers in the Republic of Nearland agree that the country will suffer a recession in the near future. At an economics conference at Nearland's premier university, four professors were having a lively discussion about how economic theory, the business cycle, and investment performance interact.

Professor Adams made two statements about the utility of consumption being an important driver of interest rates:

Statement 1: "If we believe that a recession is likely in the future, we would expect the marginal utility of consumption in the future to be lower relative to the utility of current consumption, and the inter-temporal rate of substitution to be higher than they would otherwise be if there was optimism about future economic conditions."

Statement 2: "It is my estimate that consumption in one year's time has 5% less utility than consumption in the present."

Professor Brady poured scorn on Professor Adams' second statement. He replied that predicting such precise values for abstract economic concepts such as utility was impossible. Trying to value bonds and interest rates, he argued, was much more likely to be accurate if calculations were based on more measurable macroeconomic fundamentals, such as GDP growth and inflation. Central bank policy rates, he stated, are positively correlated to current inflation and current GDP growth.

Professor Chapman attempted to mediate between Adams and Brady. He said, "In a way, you both have a point. At the top of the business cycle there is higher inflation and current GDP growth. At the same time, market participants begin to worry about future recession, which increases the marginal utility of delayed consumption. These conditions both explain a steepening of the upward-sloping yield curve."

Professor Douglas tried to move the conversation towards the business cycle and risk premiums. She presented the audience with economic data as presented in Exhibit 1, and challenged observers to calculate the equity risk premium in Nearland.

Exhibit 1: Economic Data for Nearland

Break-even rate of inflation	2.80%
Credit spread	3.10%

Risk premium for equities relative to debt 4.50%

Professor Douglas went on to discuss asset classes, which could act as a consumption hedge to guard against the upcoming recession. She invited the audience to attend her upcoming discussion seminar where she would analyze the prospects for real estate, growth stocks, and value stocks.

Question #1 - 6 of 31

Question ID: 1474026

Statement 1 made by Professor Adams is *most* likely to be:

- A) correct. 
- B) incorrect in respect of the marginal utility of consumption. 
- C) incorrect in respect of the inter-temporal rate of substitution. 

Explanation

If recession is likely, individuals become concerned that their standard of living will fall in the future, making saving more attractive. Ut, the marginal utility of delayed consumption, will be higher than normal. The inter-temporal rate of substitution, U_t / U_0 , will likewise be higher than normal.

(Module 39.1, LOS 39.c)

Question #2 - 6 of 31

Question ID: 1474027

If Statement 2 made by Professor Adams is correct, the one year real risk-free rate of return will *most* likely be closest to:

- A) 4.95%. 
- B) 5.00%. 
- C) 5.26%. 

Explanation

If the utility of delayed consumption is 5% less than consumption at present, setting the utility of present consumption as 1 implies the utility of delayed consumption is 0.95.

$$\begin{aligned}\text{intertemporal rate of substitution} = m_t &= \frac{\text{marginal utility of delayed consumption}}{\text{marginal utility of present consumption}} \\ &= \frac{0.95}{1}\end{aligned}$$

$$= 0.95$$

The risk-free rate of return can then be calculated:

$$R = \frac{1}{E(m_t)} - 1 = \frac{1}{0.95} - 1$$

$$= 5.26\%$$

(Module 39.1, LOS 39.c)

Question #3 - 6 of 31

Question ID: 1474028

Assuming the Taylor rule holds, Professor Brady's statement on the correlation of interest rates with macroeconomic fundamentals is *most* likely:

- A) correct. 
- B) incorrect in respect of the correlation of interest rates with inflation. 
- C) incorrect in respect of the correlation of interest rates with GDP growth. 

Explanation

The Taylor rule links central bank interest rates to inflation and GDP growth:

$$r = R_n + \pi + 0.5(\pi - \pi^*) + 0.5(y - y^*)$$

where:

r = central bank policy rate implied by the Taylor rule

R_n = neutral real policy interest rate

π = current inflation rate

π^* = central bank's target inflation rate

y = log of current level of GDP

y^* = log of central bank's target (sustainable) GDP

If inflation and GDP rise, central bank policy rates would be expected to rise if the Taylor rule holds.

(Module 39.1, LOS 39.d)

Question #4 - 6 of 31

Question ID: 1474029

Professor Chapman's statement about utility, inflation, GDP growth, and the yield curve is *most* likely:

- A) correct. 
- B) incorrect as the circumstances described would have no effect on the yield curve. 
- C) incorrect as the circumstances described would flatten an upward-sloping yield curve. 

Explanation

At the top of the business cycle current inflation and GDP growth are high, which increases short-term rates. Fears about future recession, however, increase marginal utility of delayed consumption, which decreases longer-term interest rates as investors become more keen to save, increasing demand for bonds and decreasing yields. These circumstances would cause a flattening of an upward-sloping yield curve, and could even result in an inverted (downward-sloping) yield curve.

(Module 39.1, LOS 39.d)

Question #5 - 6 of 31

Question ID: 1474030

The equity risk premium in Nearland is *closest* to:

A) 4.50%.



B) 7.60%.



C) 10.40%.



Explanation

The equity risk premium is the return demanded by equity investors in excess of the nominal return on a risk-free bond. It comprises a credit risk premium (credit spread) representing the risk of default on a risky bond, as well as an additional risk premium relative to risky bonds for an investment in equities.

The break-even inflation rate comprises expected inflation as well as a risk premium for uncertainty about inflation. It is included in both the overall expected return on equity and the nominal risk-free rate, so does not affect the equity risk premium.

(Module 39.2, LOS 39.h)

Question #6 - 6 of 31

Question ID: 1474031

Of the asset classes mentioned by Professor Douglas, the *most* likely to be suitable as a consumption hedge is:

A) real estate.



B) growth stocks.



C) value stocks.



Explanation

An asset suitable as a consumption hedge is one which performs relatively better in recessionary conditions than other asset classes. Equities in general are cyclical, and would be expected to perform poorly in weak economic conditions. Real estate can be considered to have both bond-like properties in that there is a predictable stream of cash flows, and equity-like properties in the uncertainty of the future value of the property. The equity-like element of real estate investments makes them unsuitable as a consumption hedge.

Growth stocks (high P/E, low dividend yields) tend to be in immature markets with high growth prospects. Value stocks (low P/E, high dividend yields, stable earnings) are commonly in established markets. A value strategy tends to perform well in recessionary conditions, while a growth strategy is more suitable for economic expansions.

(Module 39.2, LOS 39.k)

Question #7 of 31

Question ID: 1474010

Janet Grange's current one-period inter-temporal rate of substitution is 0.95. Janet is *most likely* to invest in a default-free inflation indexed one-period bond if:

- A) The bond's return is 4.89% or more.
- B) The bond's return is 5.26% or more.
- C) The bond's return is 5% or more.

**Explanation**

Real risk-free rate = $(1/E(\text{inter-temporal rate of substitution}) - 1 = (1/0.95) - 1 = 0.0526$ or 5.26%

(Module 39.1, LOS 39.c)

Question #8 of 31

Question ID: 1474021

ABC Inc. stock's price is inversely related to the business cycle; it is higher during economic downturns. Which of the following appropriately characterizes the consumption hedging property of an investment in ABC stock and the equity risk premium demanded by investors for an investment in it?

- A) Due to its desirable consumption hedging ability, an investment in ABC stock would command a higher equity risk premium.
- B) Due to its desirable consumption hedging ability, an investment in ABC stock would command a lower equity risk premium.
- C) Due to its poor consumption hedging ability, an investment in ABC stock would command a higher equity risk premium.

**Explanation**

Assets that pay off during times of scarcity provide desirable consumption hedging property. Investors will demand a lower equity risk premium on such assets.

(Module 39.2, LOS 39.h)

Question #9 of 31

Question ID: 1474032

As compared to an investment in equities, the difference in discount rate for valuation of commercial real estate is *most likely* due to:

- A) the break-even inflation rate.
- B) inflation uncertainty.
- C) lack of liquidity.



Explanation

The additional risk-premium for real estate is due to lack of liquidity.

(Module 39.2, LOS 39.k)

Question #10 of 31

Question ID: 1474011

The risk-premium for uncertainty in inflation is *most likely* to be:

- A) unrelated to the term spread.
- B) negatively related to the term spread.
- C) positively related to the term spread.



Explanation

Term spread (i.e., difference in yield between long dated government bonds and short-dated government bonds) is normally positive. Given that longer-term government bonds have a higher risk premium for uncertainty in inflation; term spread would be higher resulting in a positive relationship.

(Module 39.1, LOS 39.d)

Question #11 of 31

Question ID: 1474015

Credit spreads are *most likely* to:

- A) Rise during expansions.
- B) Rise during recessions.
- C) Rise with policy rates.



Explanation

Credit losses (probability of default and loss given default) tend to be higher during recessions and hence credit spreads are higher during recessions.

(Module 39.1, LOS 39.f)

Question #12 of 31

Question ID: 1474014

Break-even inflation rate is *most appropriately* described as the:

- A) The difference between market's expectation of the inflation rate and the risk premium for inflation uncertainty. 
- B) The difference in yields between long-dated and short-dated government bonds. 
- C) The difference in yields of non-inflation indexed and inflation indexed risk-free bonds. 

Explanation

Break-even inflation is the difference in nominal and real risk-free rates. It comprises premium for inflation and the risk premium for uncertainty in inflation.

(Module 39.1, LOS 39.e)

Question #13 of 31

Question ID: 1474003

Market values of assets are *most likely* to be affected when:

- A) New information reveals that the market's expectations about earnings were accurate. 
- B) New information reveals that the market's expectations about earnings were inaccurate. 
- C) New information confirms the market's expectations of future earnings. 

Explanation

Market values change when new information differs from expectations that are currently priced in.

(Module 39.1, LOS 39.b)

Question #14 of 31

Question ID: 1474022

Corporate earnings from which of the following sectors would be most sensitive to business cycle?

A) Consumer durable



B) Consumer non-discretionary



C) non-cyclical



Explanation

Earnings of companies in cyclical industries such as consumer durable or consumer discretionary would be more sensitive to business cycle as opposed to companies in non-cyclical industries such as consumer non-discretionary.

(Module 39.2, LOS 39.i)

Question #15 of 31

Question ID: 1474005

The marginal utility of current consumption is *most likely* higher:

A) During economic contractions.



B) During economic expansions.



C) Higher the wealth.



Explanation

Diminishing marginal utility of wealth means that an investor's marginal utility of consumption declines as wealth increases. This suggests that marginal utility of consumption is higher during periods of scarcity, such as during economic contractions.

(Module 39.1, LOS 39.c)

Question #16 of 31

Question ID: 1474019

Spreads for issuers in the consumer cyclical sector are *most likely* to:

A) Be unrelated to business cycle.



B) Decrease during economic downturns.



C) Increase during economic downturns.



Explanation

Spreads on issues from the consumer cyclical sectors tend to rise during economic downturns and fall during expansions reflecting cyclicalities in earnings of the companies in the sector.

(Module 39.2, LOS 39.g)

Question #17 of 31

Question ID: 1474012

Which of the following is *most likely* to be the shape of the yield curve during recessions?

- A) Upward sloping 
- B) Downward sloping 
- C) Flat 

Explanation

During recessions, policy rates tend to be low. Over a longer period, investor's expect inflation to be higher as the economy comes out of recession and hence longer-term rates tend to be higher resulting in an upward sloping yield curve.

(Module 39.1, LOS 39.d)

Question #18 of 31

Question ID: 1474009

The price of a zero-coupon, inflation indexed, risk-free bond that pays \$1 in one period is:

- A) Uncertain. 
- B) The expected value of the investors' inter-temporal rate of substitution between current period and one period from now. 
- C) \$1.00 

Explanation

The price of a The price of a zero-coupon, inflation indexed, risk-free bond that pays \$1 in one period is the expected value of the investors' inter-temporal rate of substitution between current period and one period from now. This value is less than \$1 as the utility of current consumption is greater than consumption in one period in the future.

(Module 39.1, LOS 39.c)

Question #19 of 31

Question ID: 1474023

Price multiples are *least likely* to increase when:

- A) Inflation expectations decline. 
- B) Equity risk premium increases. 

C) Earnings growth increases.



Explanation

Price multiples are positively related to earnings growth and negatively related to each of the components of the discount rate (i.e., real rate, inflation premium, equity risk premium).

(Module 39.2, LOS 39.j)

Question #20 of 31

Question ID: 1474013

If the market expects inflation to decrease over the next few years but the uncertainty about inflation was increasing, the break-even inflation rate is *most likely* to:

A) Increase.



B) Be uncertain.



C) Decrease.



Explanation

BEI = expected inflation + risk premium for uncertainty about inflation. While inflation is expected to decrease, the higher inflation uncertainty increases the risk premium. Hence the overall impact is uncertain.

(Module 39.1, LOS 39.e)

Question #21 of 31

Question ID: 1474002

Market values of assets are *most likely* to be affected when either:

A) Risk free interest rates, risk premiums, timing and/or magnitude of expected cash flows change.



B) Real risk-free rates, risk premiums, timing/magnitude of expected cash flows change.



C) Real risk-free rates, inflation premium, timing/magnitude of expected cash flows change.



Explanation

Market values of assets are affected when the expected cash flows or discount rate changes. The discount rate can change either due to changes in risk-free rate or due to changes in risk premiums.

(Module 39.1, LOS 39.a)

Question #22 of 31

Question ID: 1474017

Jeff Dentmat is expecting overall credit spreads to narrow over the next few years. Which of the following conclusions can Dentmat most appropriately make?

- A) Government bonds will outperform low-rated corporate bonds. 
- B) Risky bonds will outperform risk-free bonds. 
- C) High rated corporate bonds will outperform low-rated corporate bonds. 

Explanation

When credit spreads narrow, lower rated bonds outperform higher rated bonds.

(Module 39.1, LOS 39.f)

Question #23 of 31

Question ID: 1474007

A high default-free interest rate is *most likely* to be associated with:

- A) expectations of higher income in the future. 
- B) investors attaching high utility to future consumption relative to current consumption. 
- C) expectations of goods and services being less available in the future relative to today. 

Explanation

An increase in real GDP growth means that more goods and services will be available in the future relative to today. Investors will be less willing to substitute across time, leading to more borrowing and less saving. This leads to an increase in the real default-free interest rate.

(Module 39.1, LOS 39.c)

Question #24 of 31

Question ID: 1474018

Differences in credit spreads across sectors are *least likely* due to:

- A) Differences in leverage typical of the sector. 
- B) Differences in ratings. 
- C) Differences in services and products that an industry produces. 

Explanation

Differences in credit spreads across sectors is related to differences in products/services the sector produces and leverage typically used in the sector.

(Module 39.2, LOS 39.g)

Question #25 of 31

Question ID: 1474020

Which of the following assets provides a most effective hedge against bad consumption outcomes?

- A) Equity 
- B) Risk-free bonds. 
- C) Real estate 

Explanation

Risk-free bonds (especially long maturity bonds) provide an effective hedge against bad consumption outcomes. Equity prices and real estate values tend to be positively related to the state of the economy and hence do not provide good hedges against bad consumption outcomes.

(Module 39.2, LOS 39.h)

Question #26 of 31

Question ID: 1474001

Which of the following is *least likely* to explain a decline in the S&P 500 index:

- A) A decrease in expectations about corporate earnings. 
- B) An increase in Treasury yields. 
- C) A decline in expected inflation. 

Explanation

Equity market prices are positively related to expected earnings/cash flows and negatively related to discount rate. Discount rate is positively related to inflation expectations and treasury yields (risk-free rate).

(Module 39.1, LOS 39.a)

Question #27 of 31

Question ID: 1474006

Rapidly developing economies like India and China have high GDP growth rates and therefore are *most likely* to have a:

- A) Low real rate, high inter-temporal rate of substitution and a low rate of current borrowing by investors. 
- B) High real rate, low inter-temporal rate of substitution and a high rate of current borrowing by investors. 
- C) High real rate, low inter-temporal rate of substitution and a low rate of current consumption. 

Explanation

High GDP growth leads to higher future expected incomes and therefore high rate of current consumption (low savings, high borrowings, and an increase in the real default-free rate of interest) and a low inter-temporal rate of substitution.

(Module 39.1, LOS 39.c)

Question #28 of 31

Question ID: 1474008

The real rate of return is *most likely* higher:

- A) Higher the utility investors attach to future consumption relative to current consumption. 
- B) Higher the expectations of lower income in the future. 
- C) Lower the utility investors attach to future consumption relative to current consumption. 

Explanation

Real rate of return is higher, higher the utility of current consumption relative to future consumption. If investors expect lower incomes in the future, the utility of future consumption relative to current consumption will be higher and real rate will be lower.

(Module 39.1, LOS 39.c)

Question #29 of 31

Question ID: 1474004

Investors are *least likely* to increase their savings rate when:

- A) Uncertainty about their future income decreases.
- B) Uncertainty about their future income increases.
- C) Expected rates of returns increase.



Explanation

Investors would increase their savings rate when uncertainty about future income increases and/or expected rates of return increase.

(Module 39.1, LOS 39.c)

Question #30 of 31

Question ID: 1474024

Value stocks are *most likely* to be characterized by:

- A) Immature markets.
- B) High earnings growth.
- C) Low price multiples.



Explanation

Value stocks tend to have low price multiples, high dividend yield and tend to be in mature industries with low earnings growth.

(Module 39.2, LOS 39.j)

Question #31 of 31

Question ID: 1474016

The break-even inflation rate is expected to be 2% over the next year. What is the credit spread for a 2% annual pay corporate bond maturing in one year with a market price of \$96.91 (\$100 par) if the real risk-free rate of return over the next year is 1%?

- A) 0.19%
- B) 2.00%
- C) 2.25%



Explanation

The YTM on the corporate bond is $(102/96.91) - 1 = 5.25\%$.

$$\text{Credit spread} = \text{Yield} - \text{BEI} - R = 5.25\% - 2\% - 1\% = 2.25\%$$

(Module 39.1, LOS 39.f)